WALCHAND COLLEGE OF ENGINEERING, SANGLI

WALCHAND LINUX USERS' GROUP

OPEN SOURCE DAY

EXCITING PRIZES

LIMITED SEATS

S1

- Introduction to GIT
- GIT Commands
- GIT Branching
- GITHub Contribution

REGISTER HERE

S2

- Introduction to CyberSecurity
- Various CyberSecurity Tools
- Security Precautions
- Detection and Solutions

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TABLE OF CONTENTS

- Open Source Software
- Version Control System
- Git Architecture

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OPEN SOURCE

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WHAT IS OPEN SOURCE?

- Source Code: A set of instructions that tells a computer how a program works.
- Open Source: refers to programs whose source code is publicly available for anyone to view and improve

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WHY OPEN SOURCE?

- Free and accessible code
- Collaborative development
- Transparency and security
- Flexibility and customization

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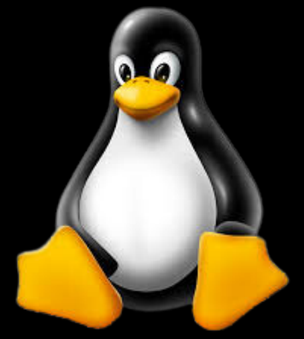
EXAMPLES OF OPEN SOURCE SOFTWARE



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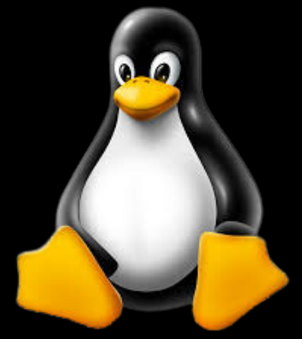
EXAMPLES OF OPEN SOURCE SOFTWARE

VLC

VS Code



Linux

Blender

Git

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VERSION CONTROL SYSTEM

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VERSION CONTROL SYSTEM

- Manages multiple versions
- Audits changes
- Simplifies code integration
- Backs up your work
- Helps in error handling

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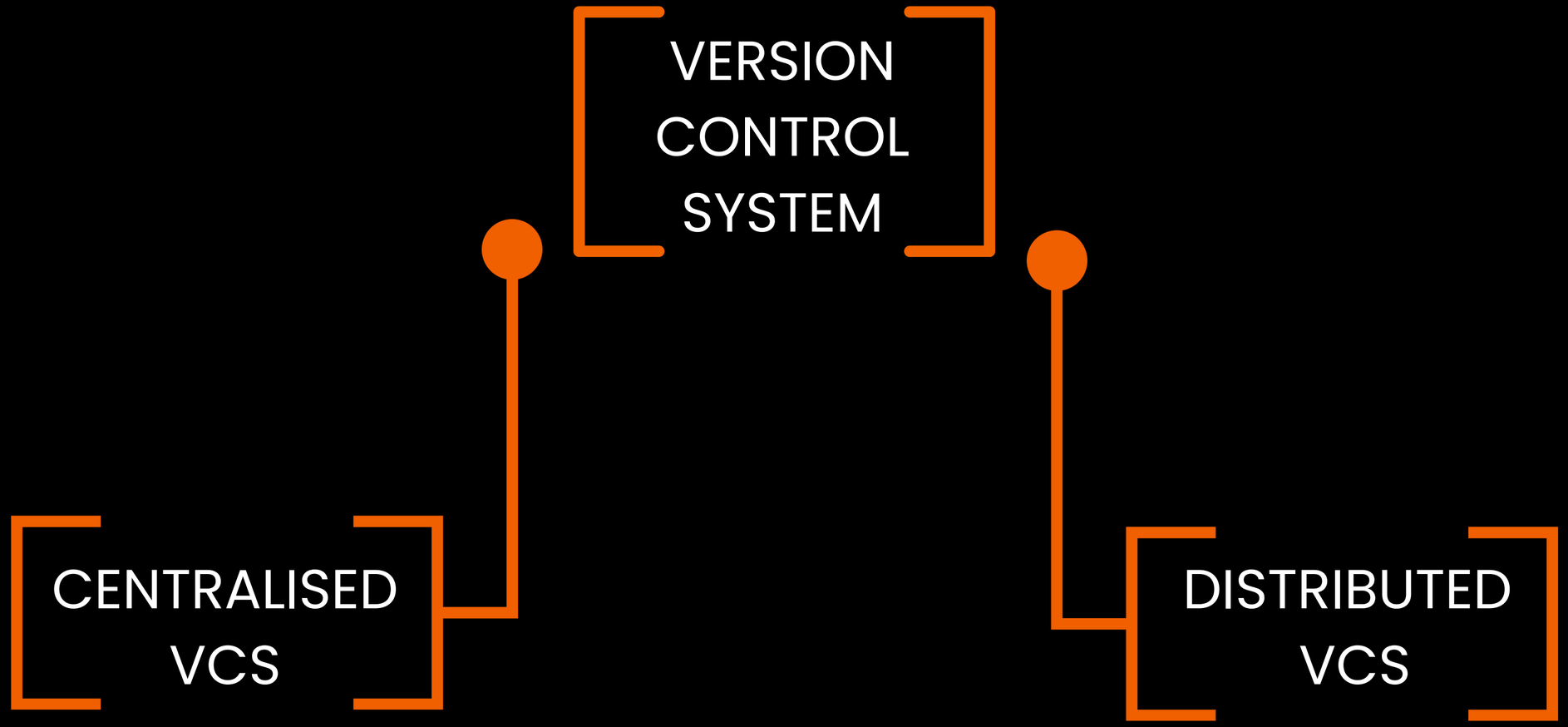
TYPES OF VERSION CONTROL SYSTEM



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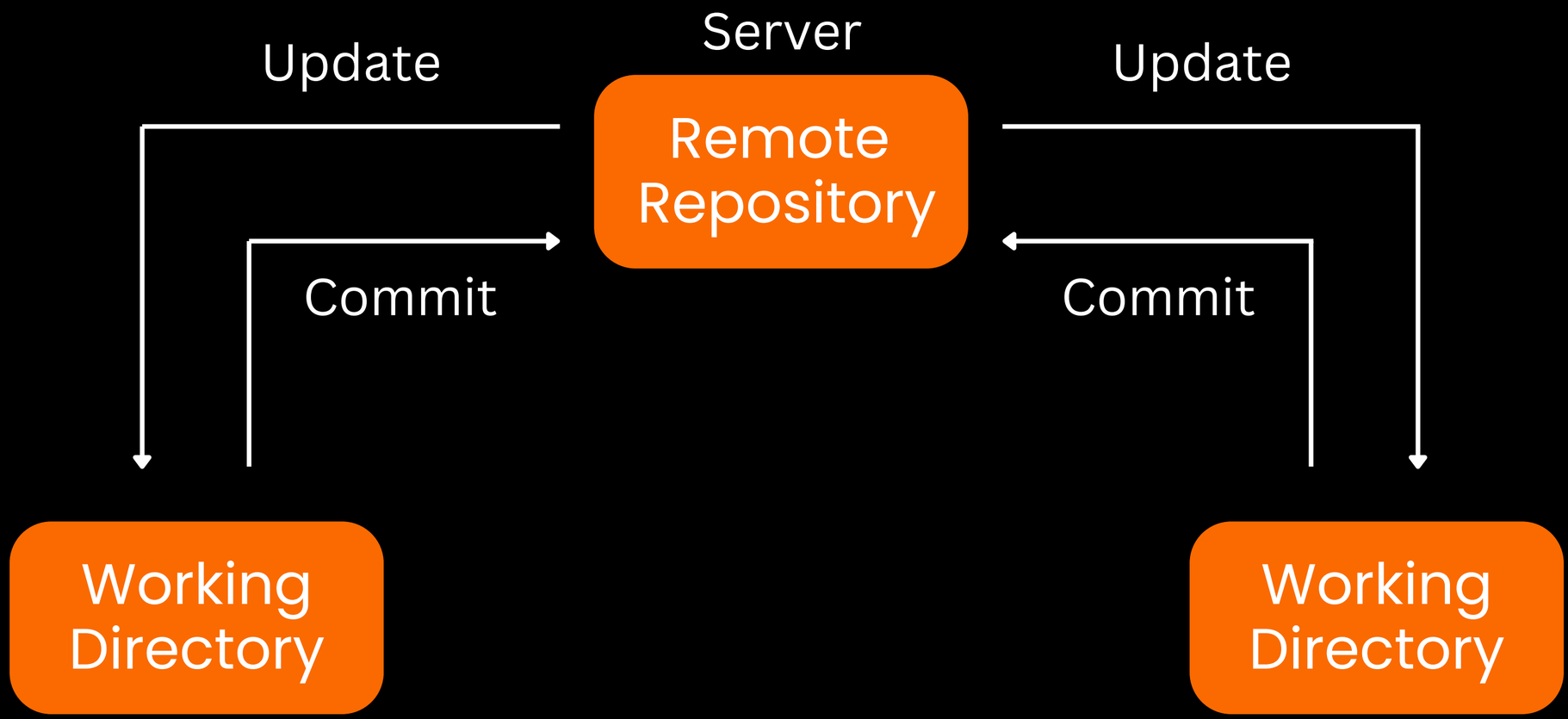
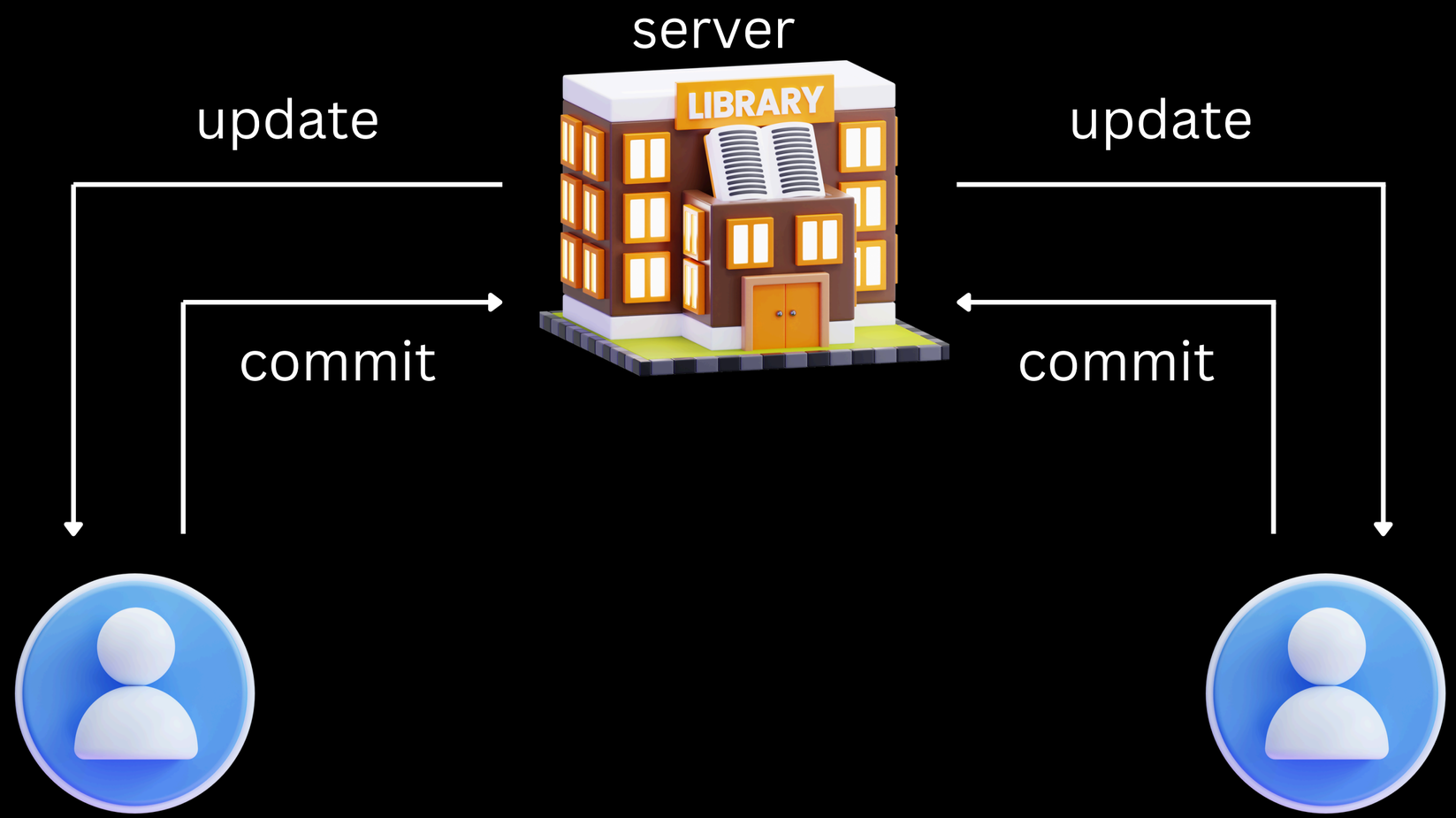
CENTRALISED VCS



CENTRALISED VCS



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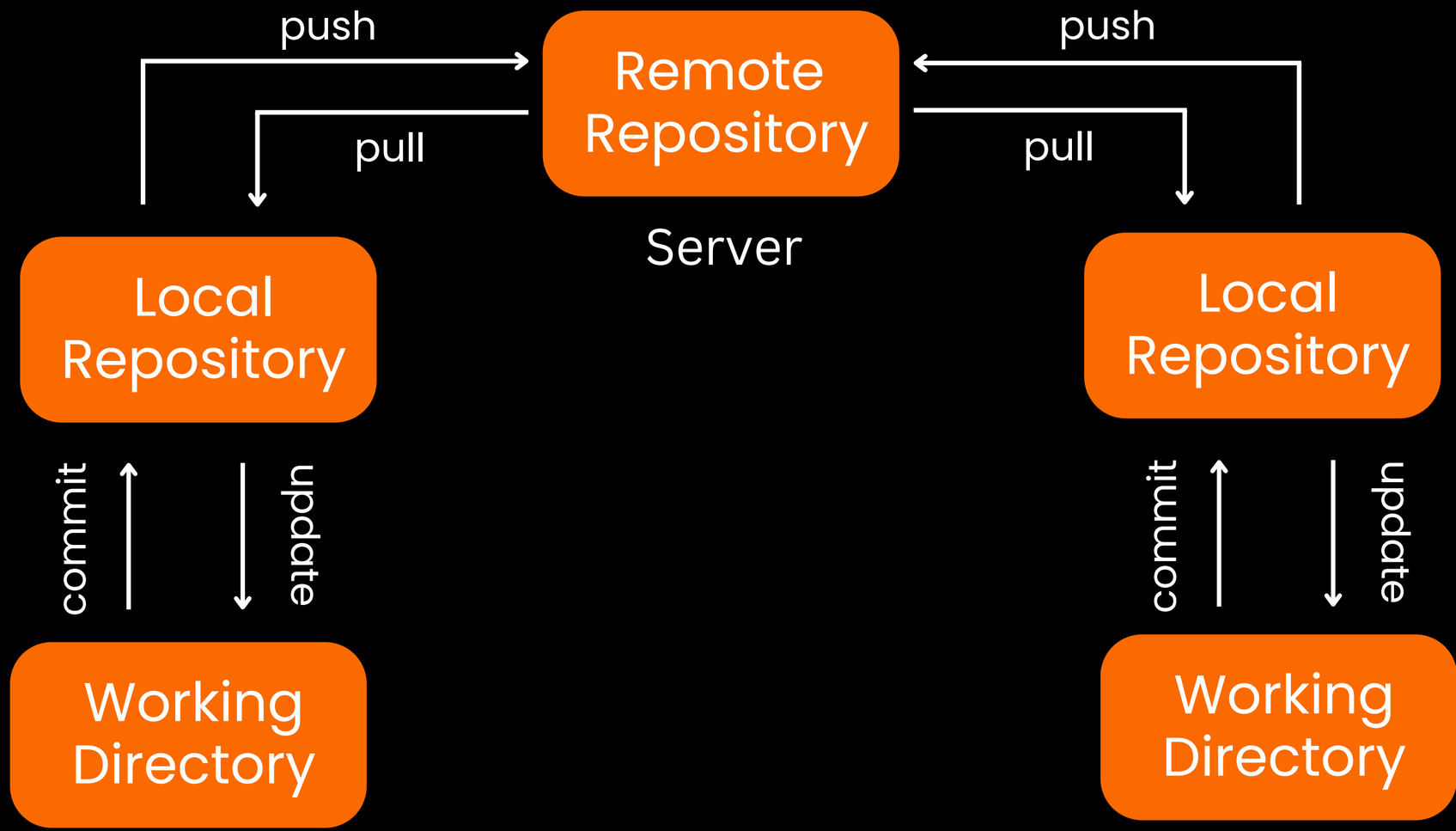
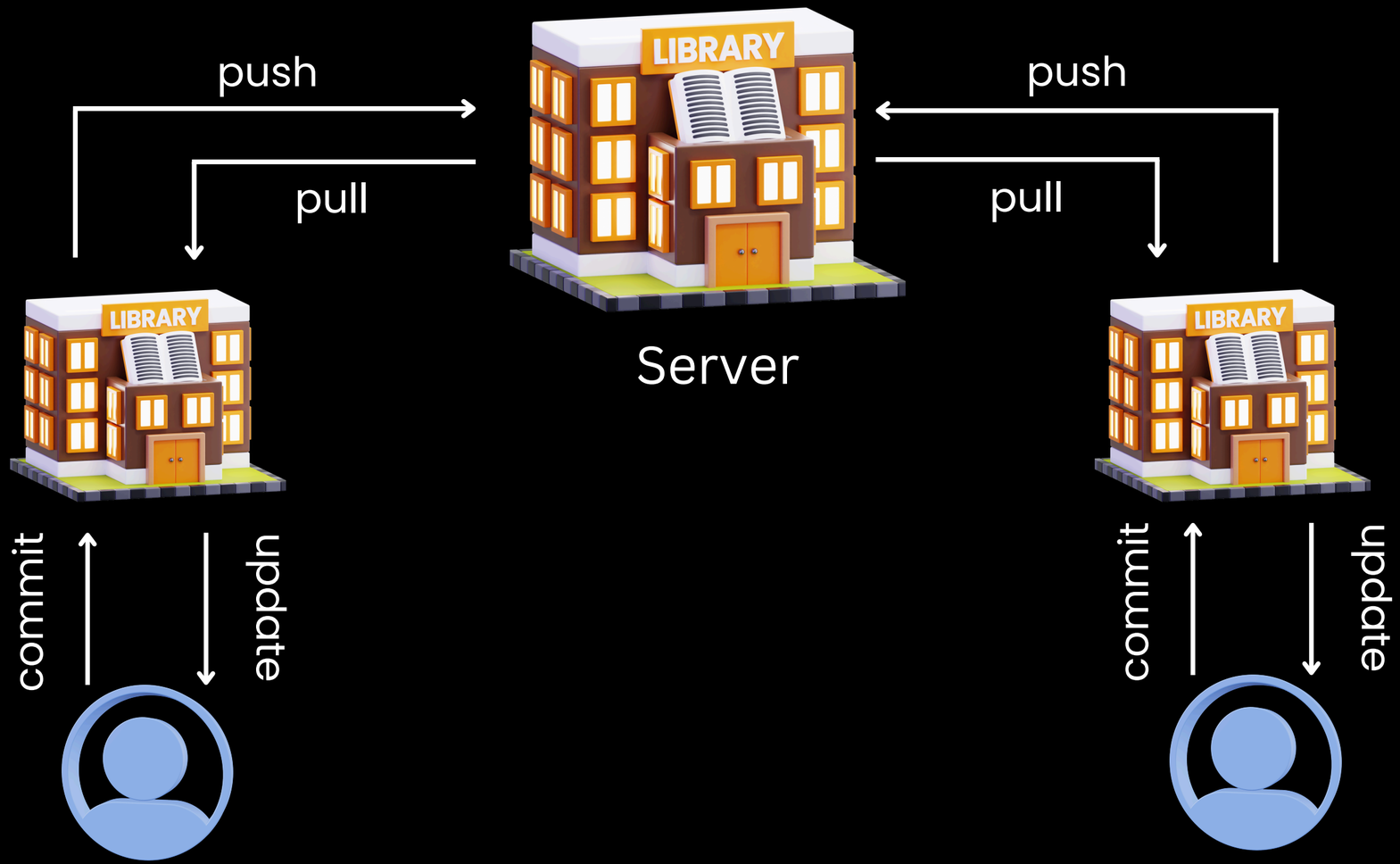
CENTRALISED VCS

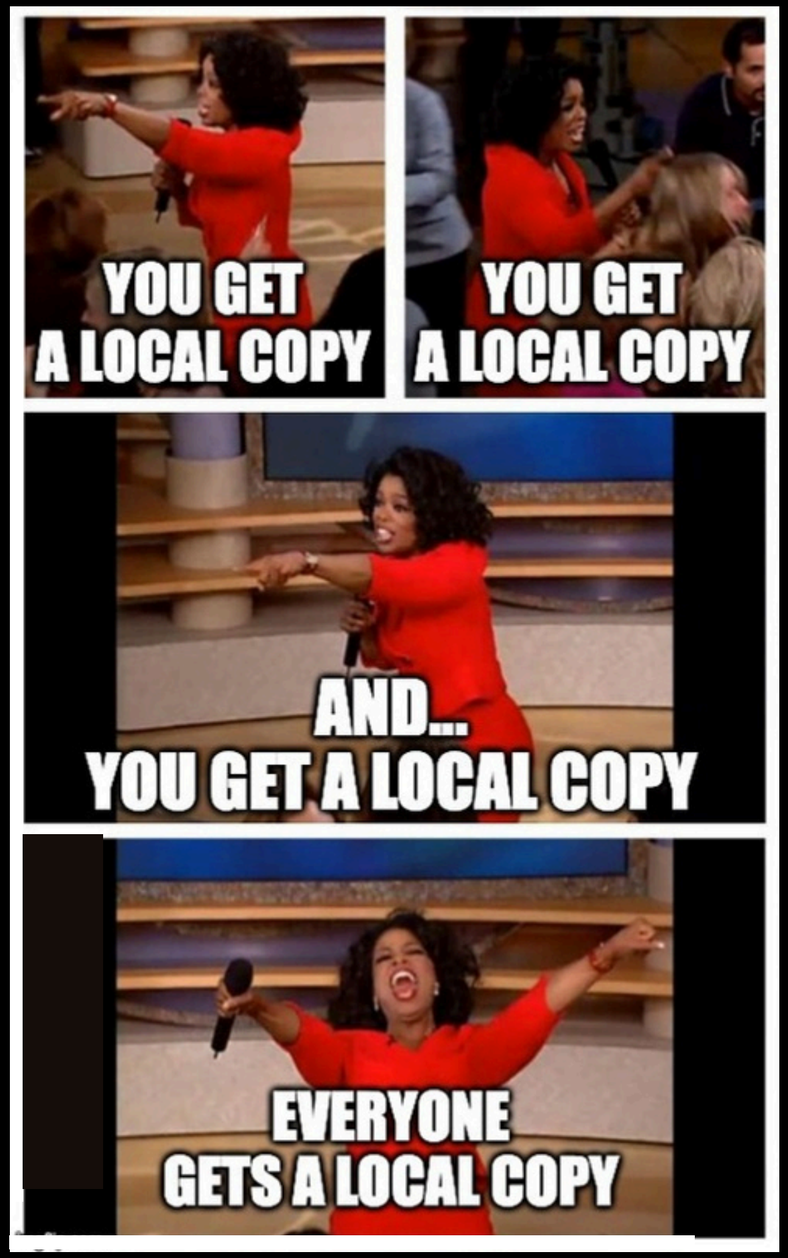
- Uses a single, central repository
- Offers limited or no offline access
- Requires a constant connection to the central server
- e.g. Subversion(SVC), Perforce

DISTRIBUTED VCS



DISTRIBUTED VCS




YOU GET
A LOCAL COPY
YOU GET
A LOCAL COPY
AND...
YOU GET A LOCAL COPY
EVERYONE
GETS A LOCAL COPY

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DISTRIBUTED VCS

- Each user has a full copy of the repository
- Full offline access to the project's history
- Users can work locally even if the central server fails
- E.g. Mercurial, Git

ARE YOU GOING TO SLEEP?
YES I AM. NOW SHUT UP.
HOW DOES GIT KEEP TRACK OF ITS VERSIONS?

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DO YOU KNOW HIM?

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DO YOU KNOW HIM?

-Linus Torvalds

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GIT HISTORY

- Created by Linus Torvalds in 2005
- Replaced BitKeeper
- Speed, security and distributed development
- Most widely used version control system

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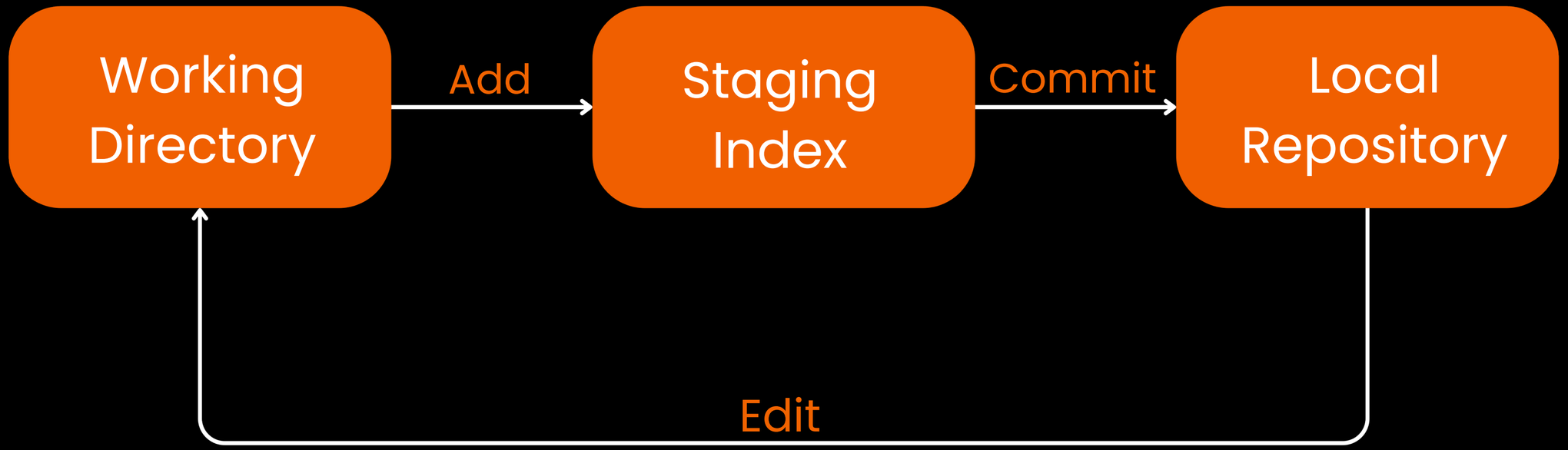
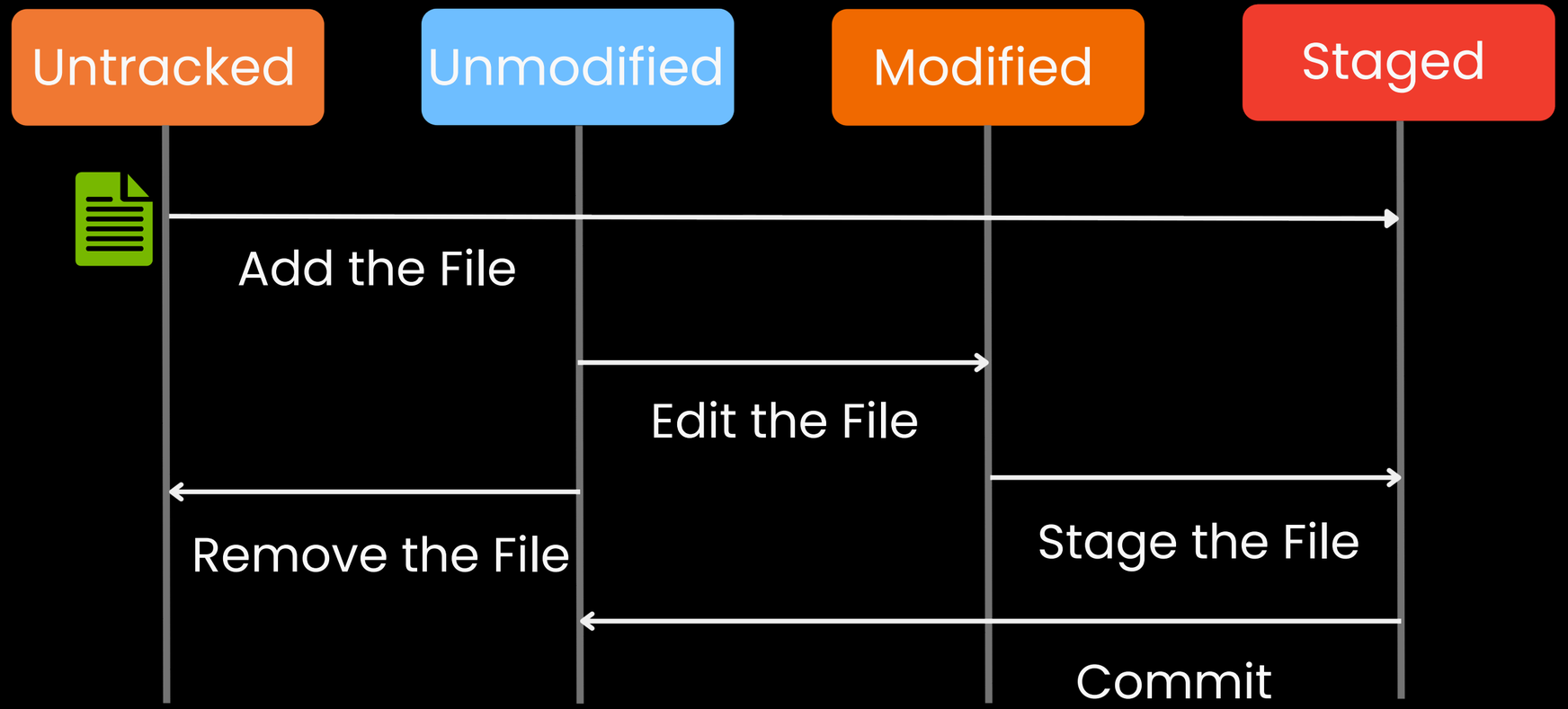
GIT ARCHITECTURE





File in VS code:

File in Git:

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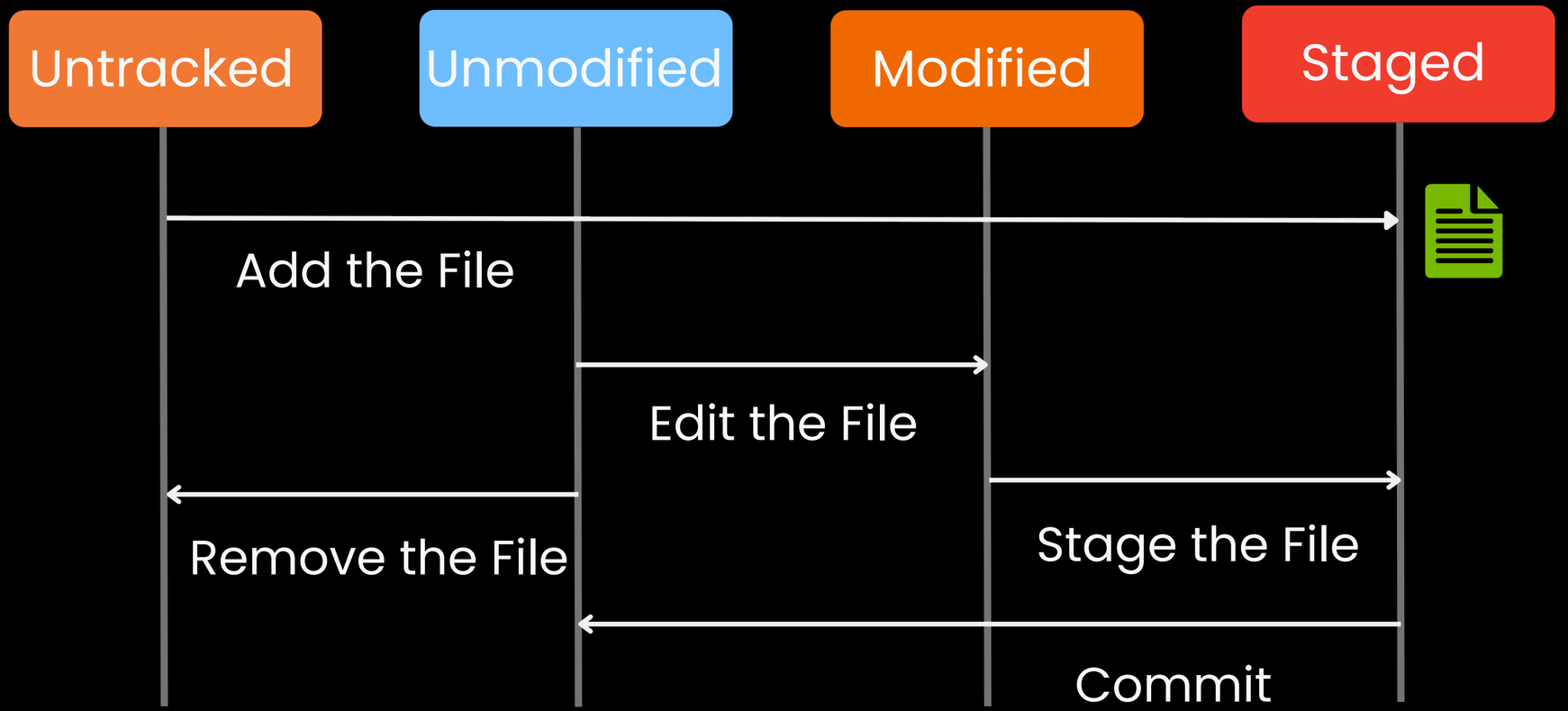
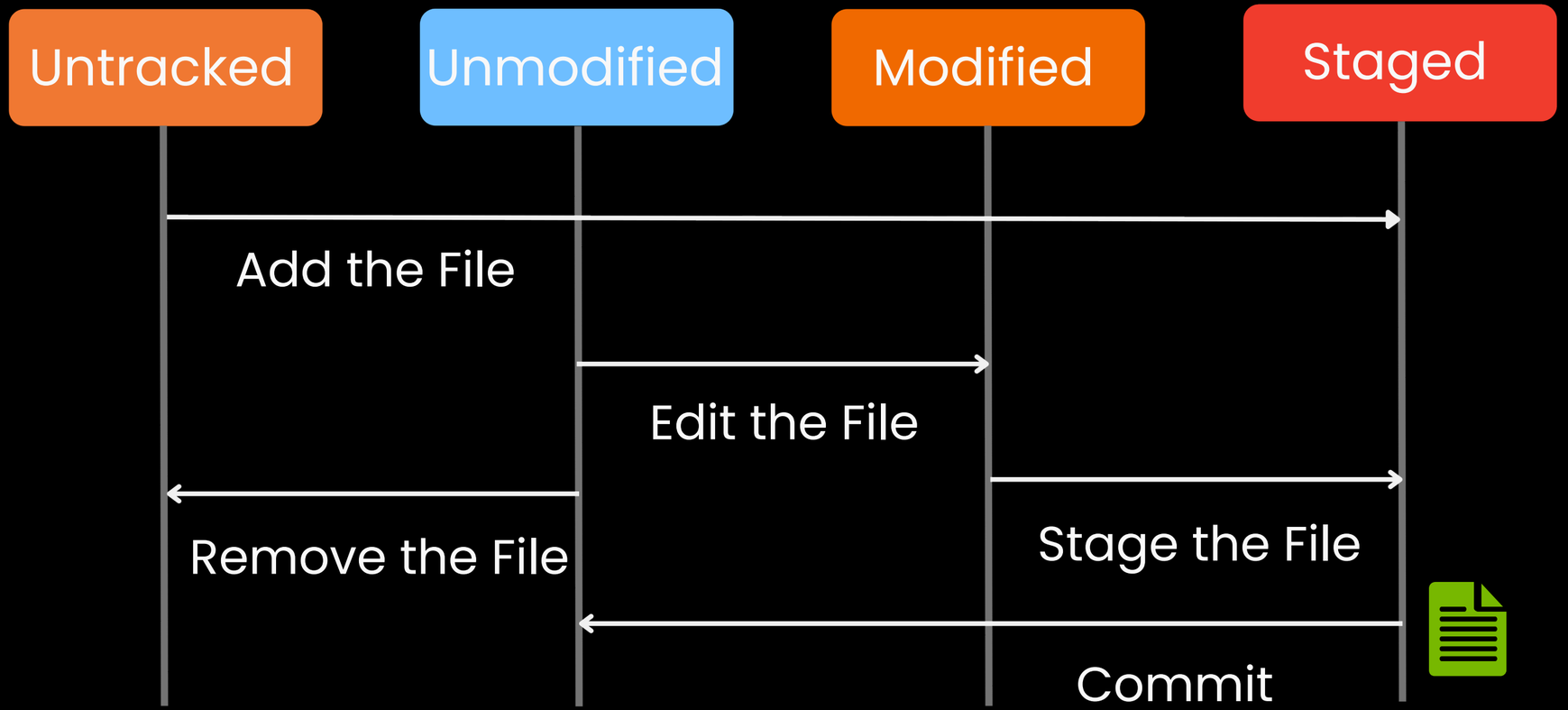


File in VS code:

File in Git:

Git knows about it!!



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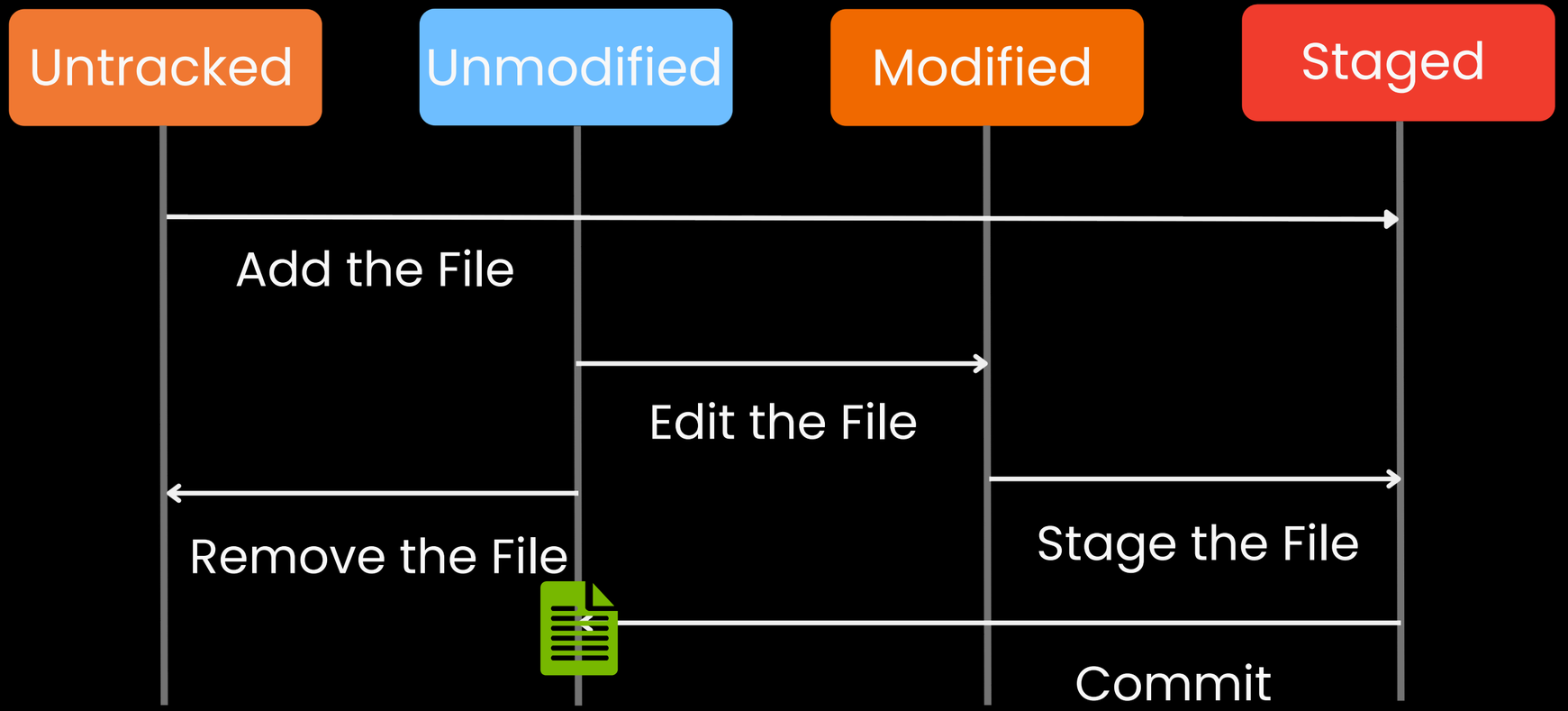

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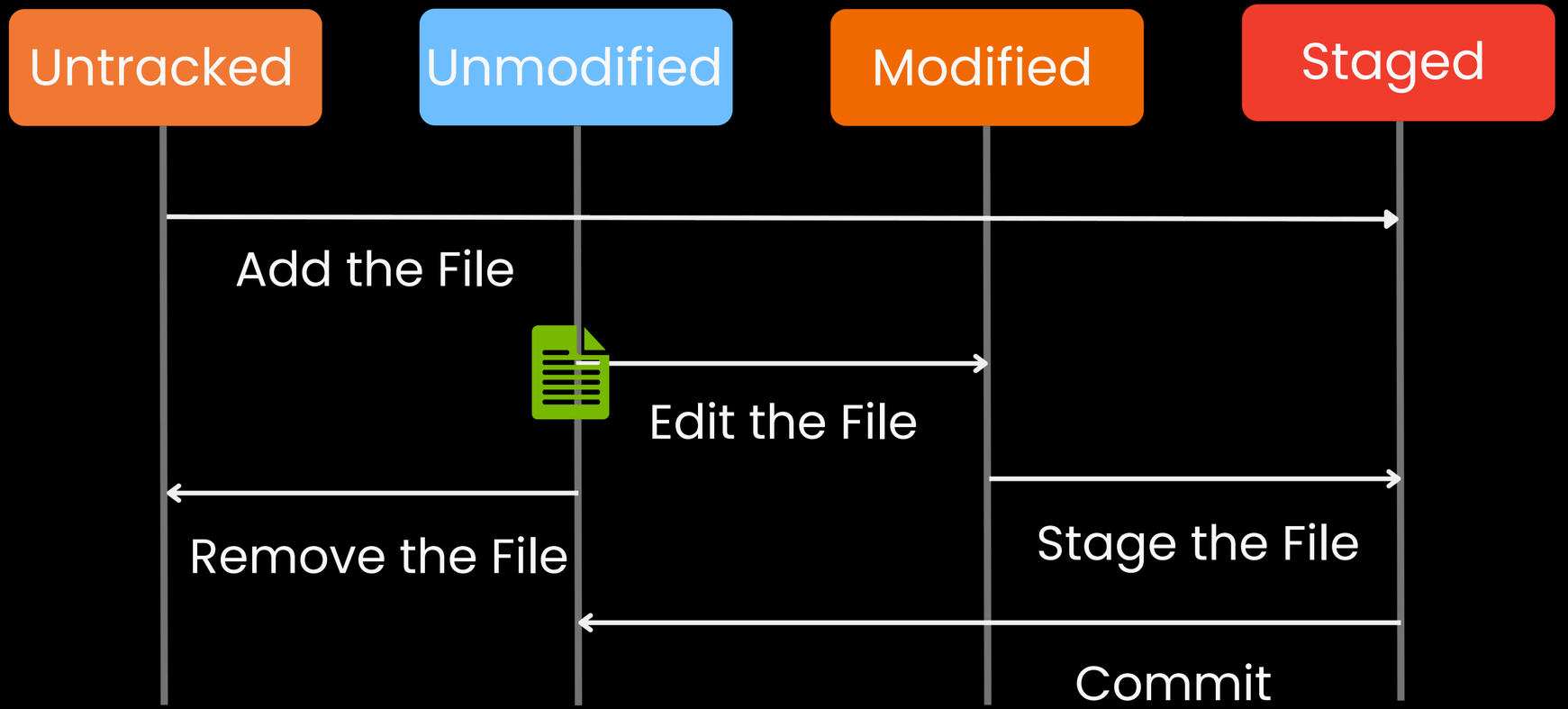
File in Git:



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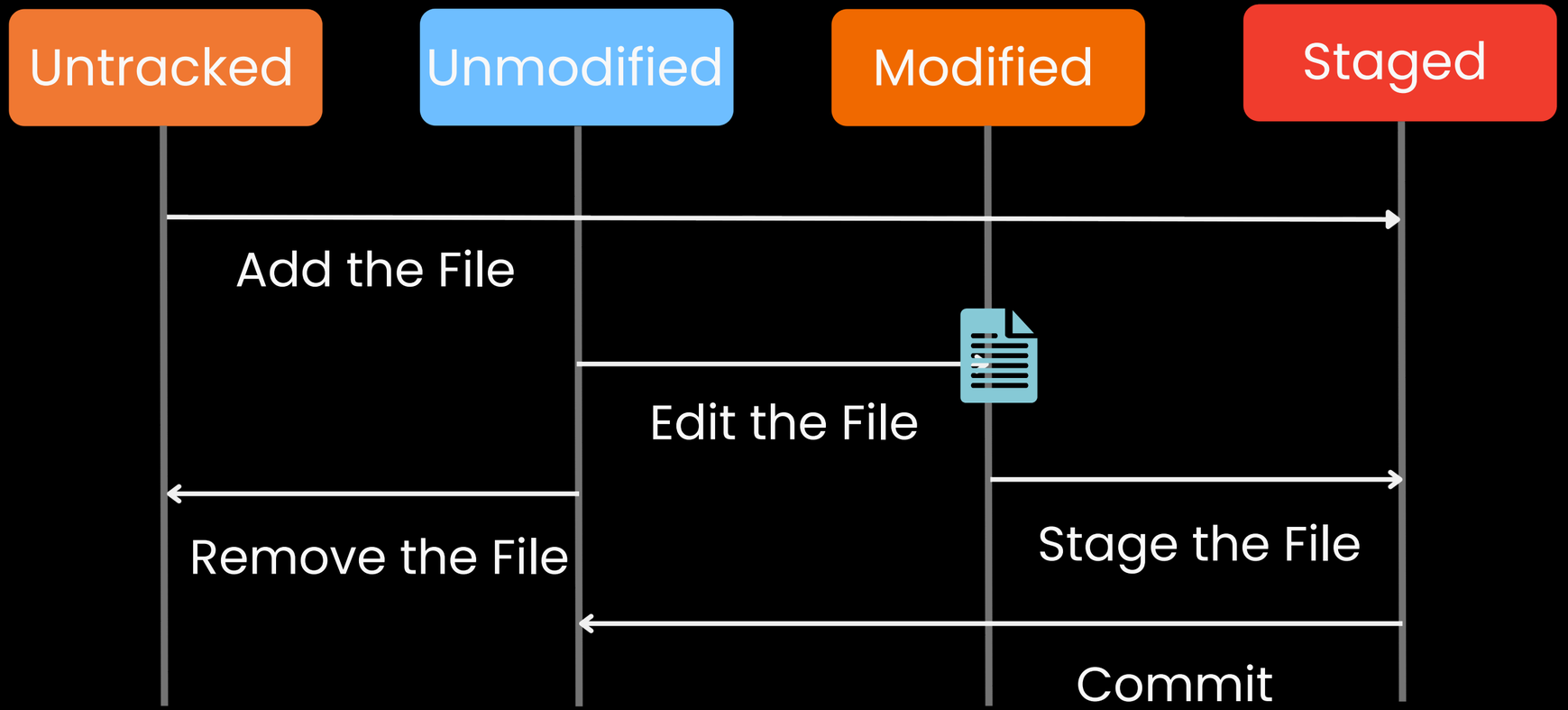

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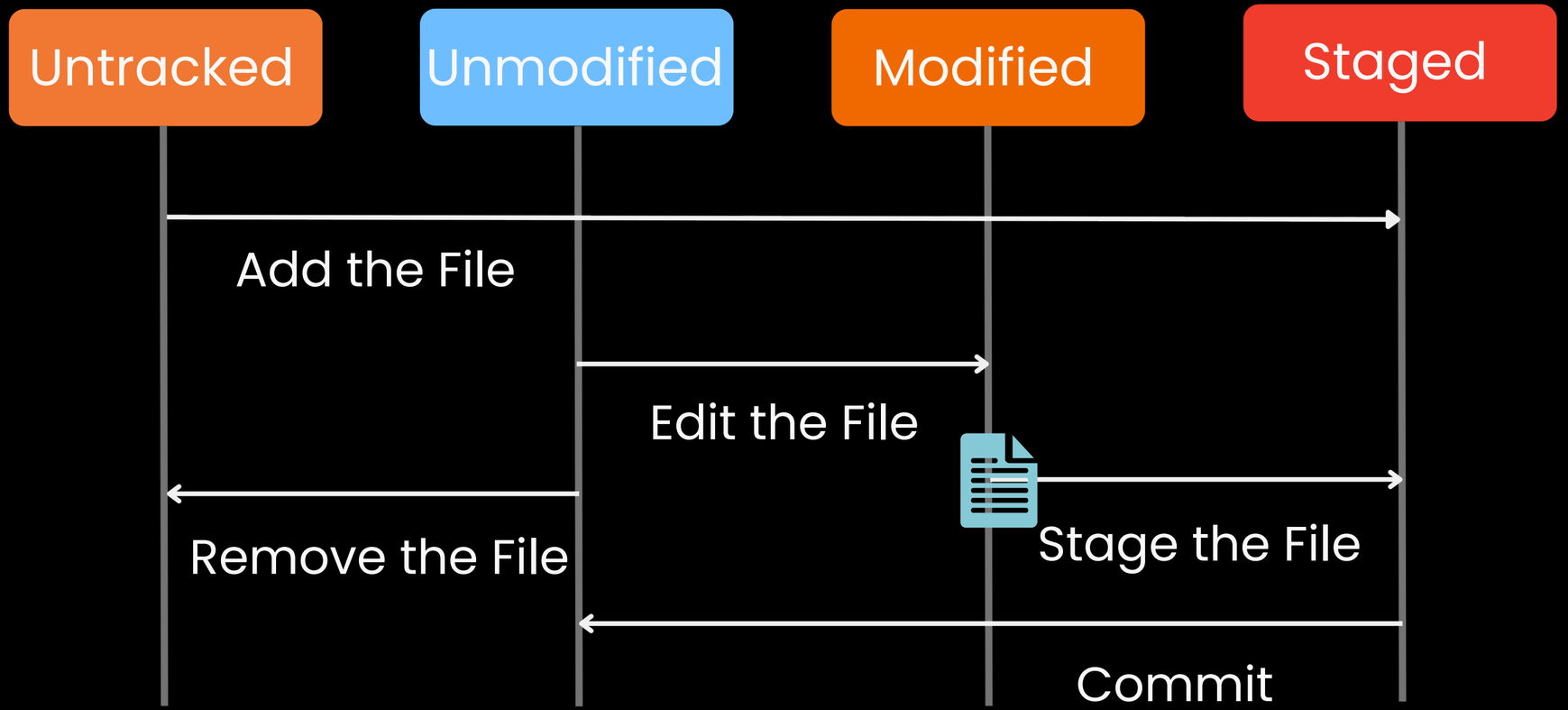

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File in Git:

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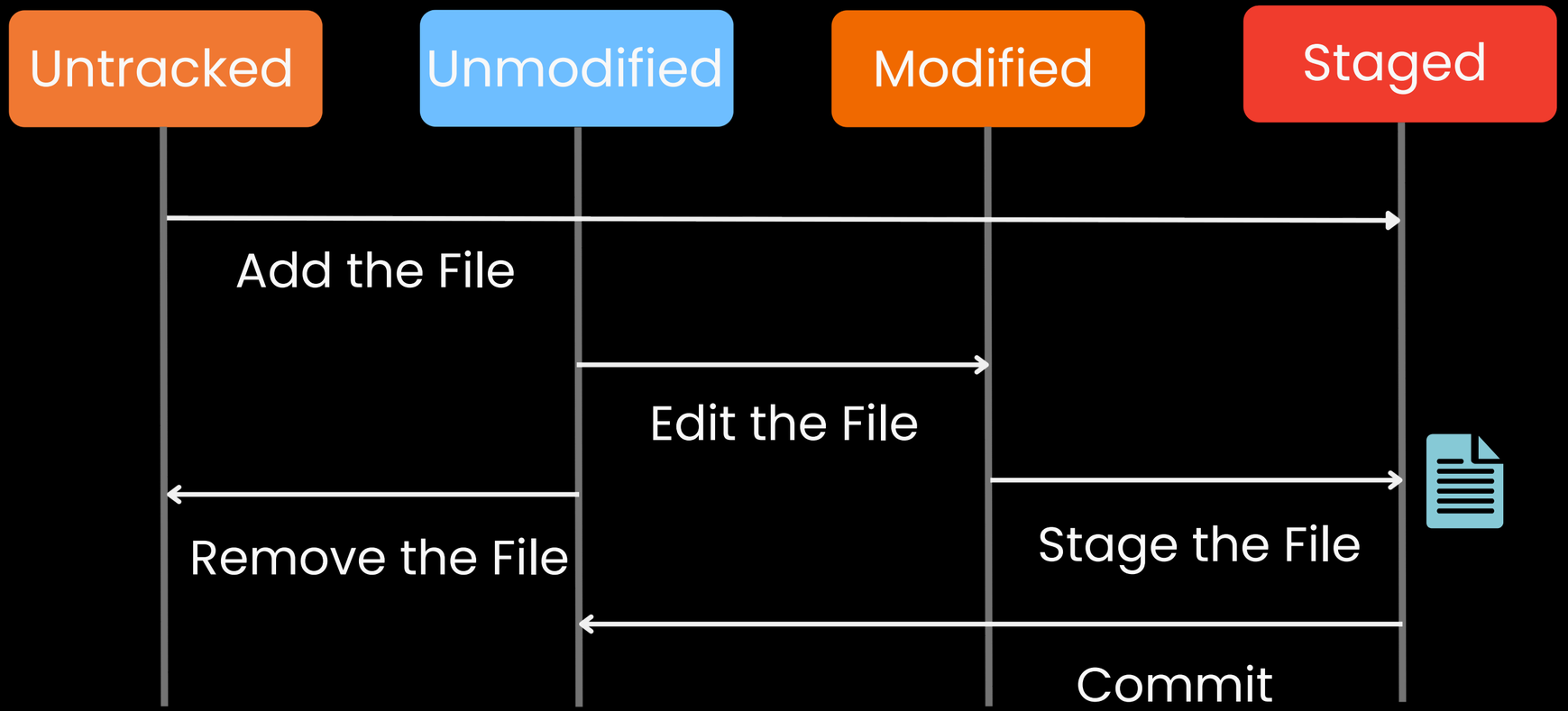
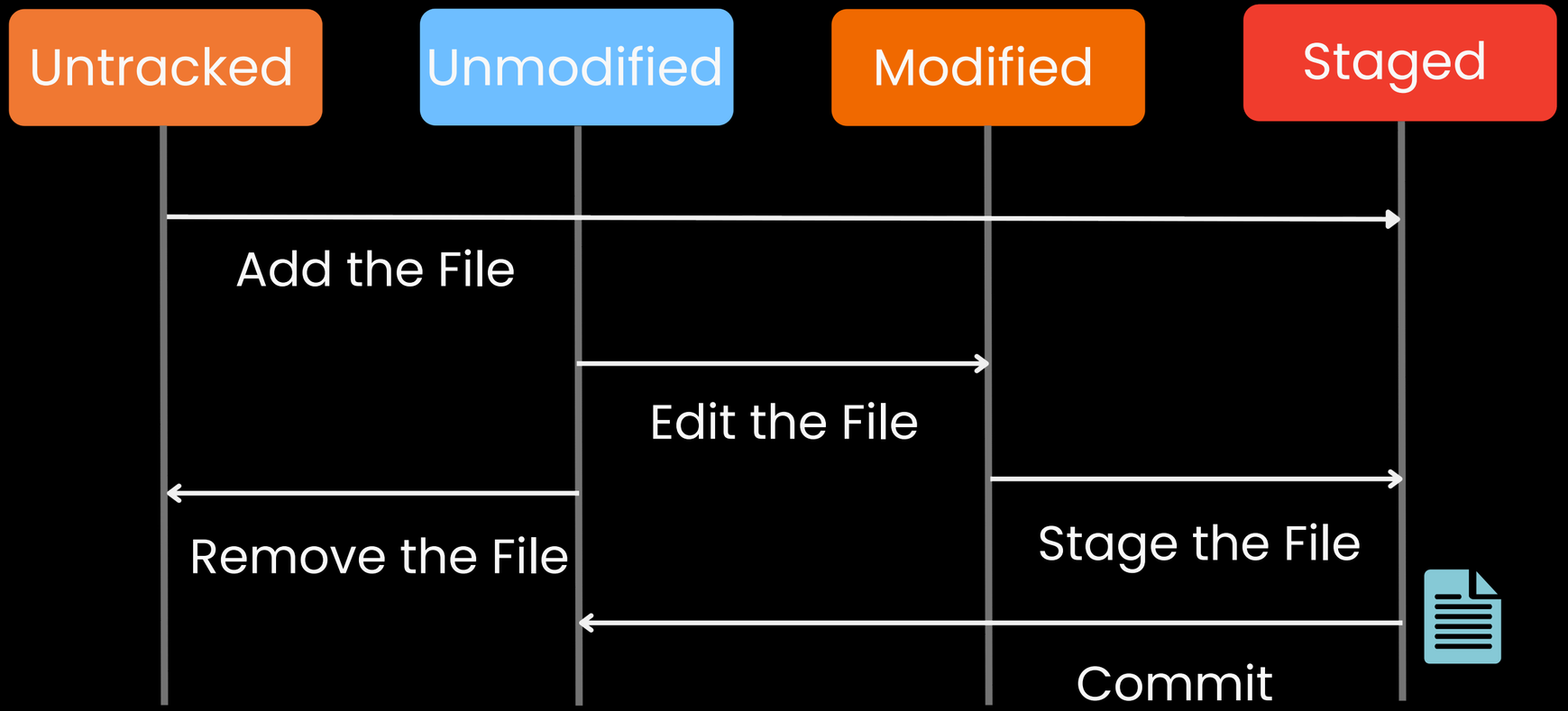


File in VS code:

File in Git:

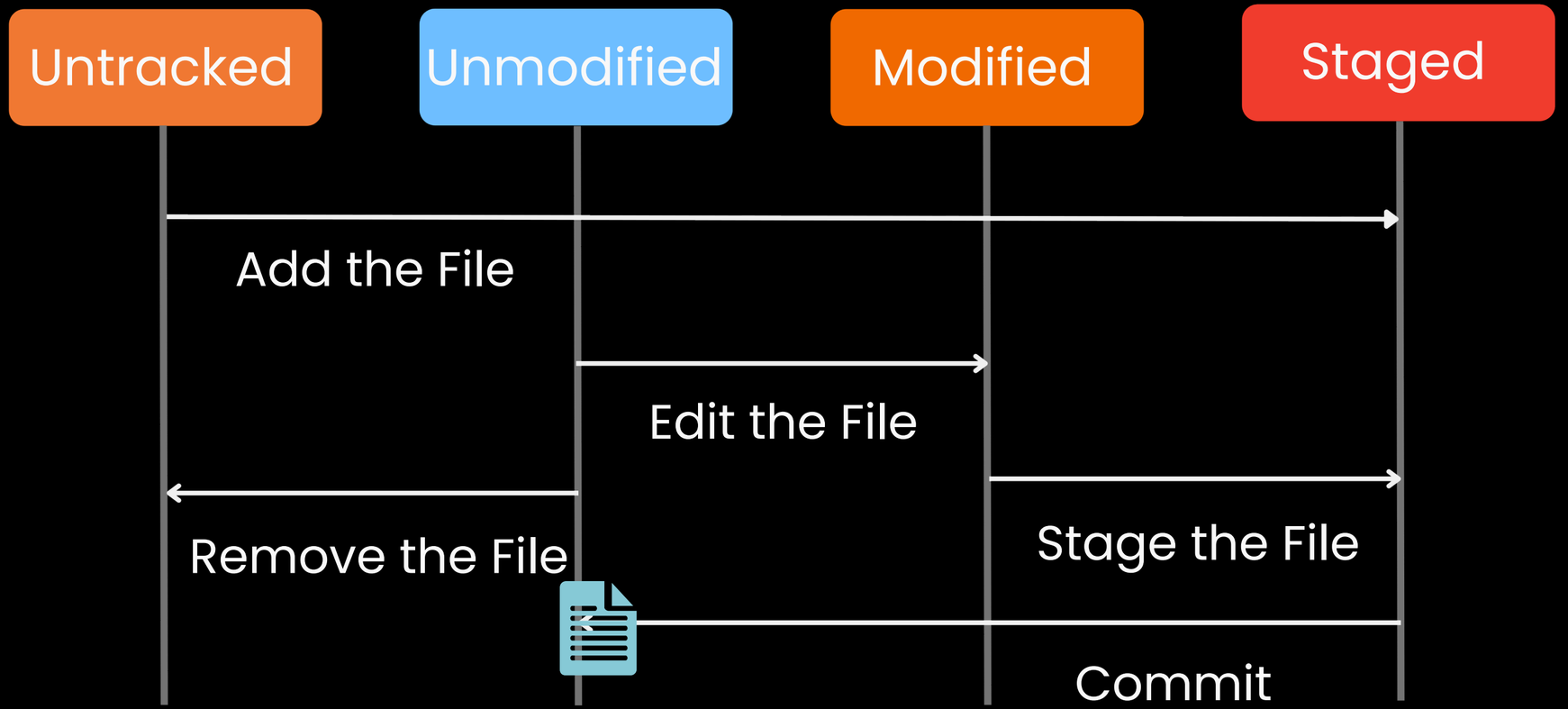
Git knows about it!!



File in VS code:

File in Git:

Git knows about it!!



File in VS code:

File in Git:

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GIT	GITHUB
Distributed VCS	Cloud based host
Installed on a local machine	Accessed on a web browser
Version control and tracking changes	Storage and collaboration
Used offline	Requires internet
Open Source software	Closed Source Software

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BASIC LINUX COMMANDS

- pwd: Prints the working directory

```
$ pwd
```

- ls: Lists the files in a directory

```
$ ls
```

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BASIC LINUX COMMANDS

- mkdir: Creates an empty directory

```
$ mkdir <directory_name>
```

- cd: Changes directory

```
$ cd <directory_name>
```

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BASIC LINUX COMMANDS

- touch: Creates a file

```
$ touch <file_name>
```

- cat: Displays the file content

```
$ cat <file_name>
```

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TABLE OF CONTENTS

- Installing & Configuring Git
- Git Commands
- Git Branching & Merging
- Advance Git Commands
- Open Source Contribution

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INSTALLING GIT

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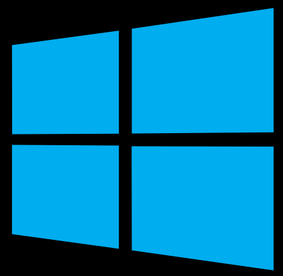
- Linux:

```
$ sudo apt install git
```

- Windows:

https://git-scm.com/install/windows



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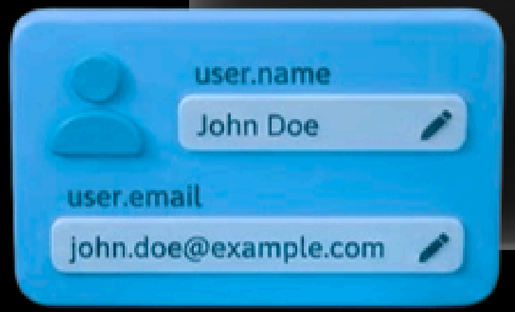
CONFIGURING GIT



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- Set Username:

```
$ git config --global user.name "your-name"
```

- Set Email:

```
$ git config --global user.email "your-email"
```

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BASIC GIT COMMANDS

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- Initializes a Git repository

```
$ git init
```

- Checks the status of the Git repository

```
$ git status
```

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- Adds a particular file to the staging area

```
$ git add <file_name>
```

- Adds all the files to the staging area

```
$ git add .
```

- Commits your changes to the local repository with a message

```
$ git commit -m <Commit Message>
```

- Shows a list of commit history

```
$ git log
```

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CONTROLLING WHAT GIT TRACKS

- Git uses a special file called .gitignore
- Specifies intentionally files or folders to ignore
- Keeps the repository clean and manageable
- node modules, .env, *.log, .txt

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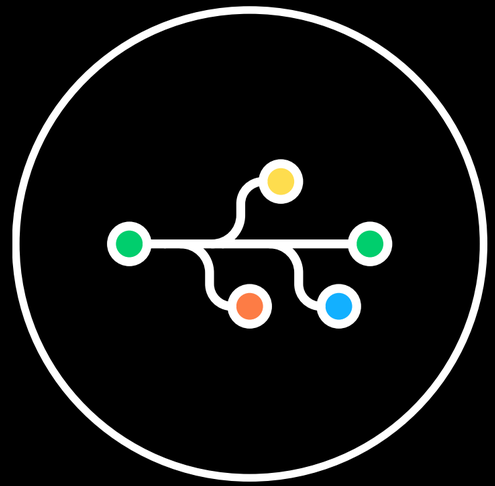
GIT BRANCHING

BRANCHING COMMANDS

- Lists all local branches

```
$ git branch
```



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- Creates a new branch

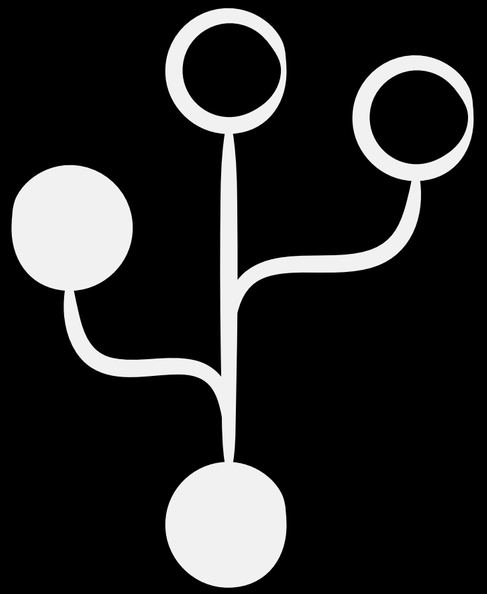
```
$ git branch <branch_name>
```

- Switches to an existing branch

```
$ git checkout <branch_name>
```

OR

```
$ git switch <branch_name>
```



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- Creates and switches to a new branch

```
$ git checkout -b <branch_name>
```

- Compares differences between two branches

```
$ git diff <branch1> <branch2>
```

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- Renames a branch

```
$ git branch -m <old_name> <new_name>
```

- Deletes a branch

```
$ git branch -d <branch_name>
```

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MERGING COMMANDS

- Merges another branch into current branch

```
$ git merge <branch_name>
```

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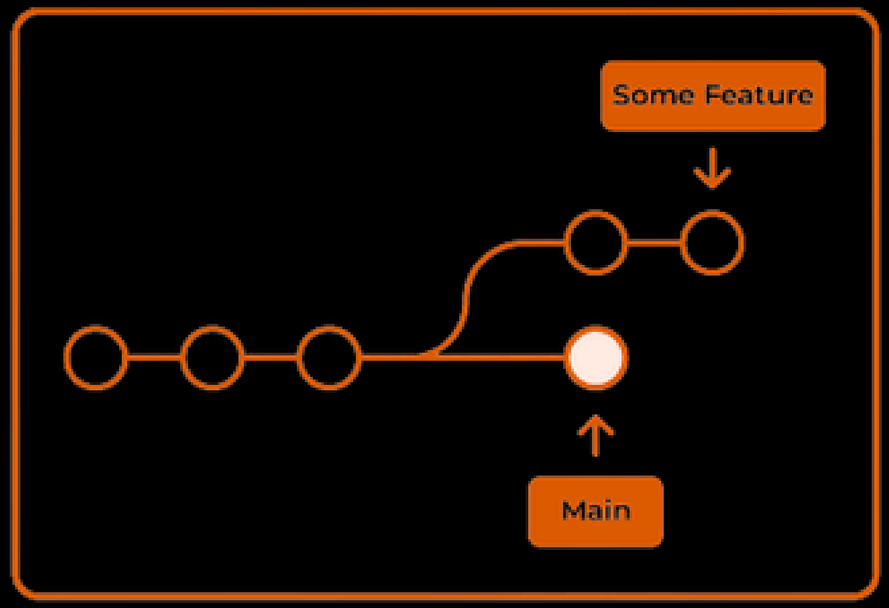
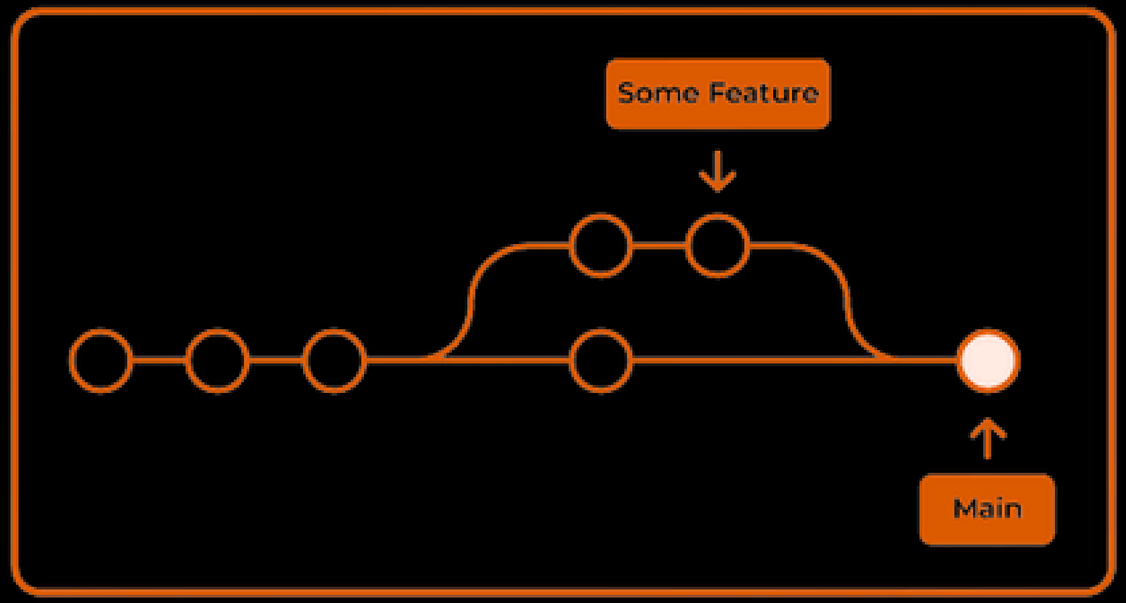
Before Merging



After a 3-way Merge



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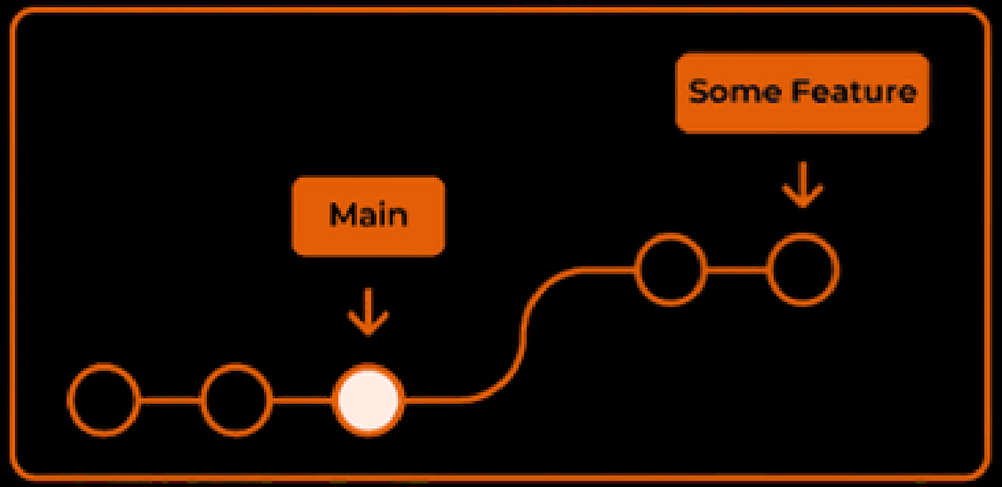
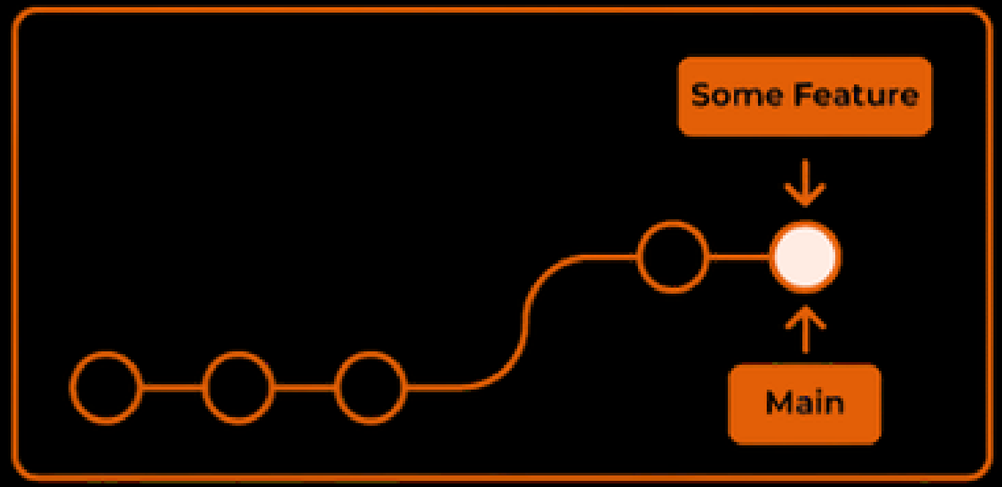
Before Merging



After a Fast-Forward Merge



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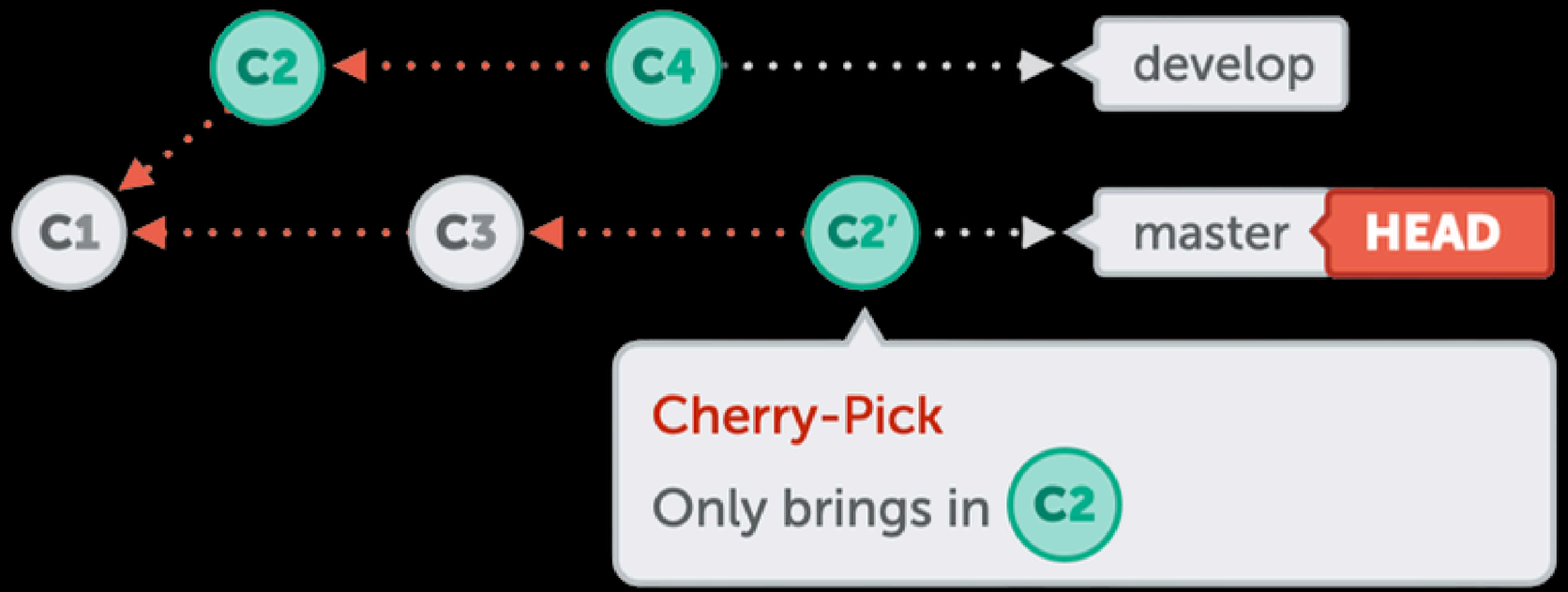
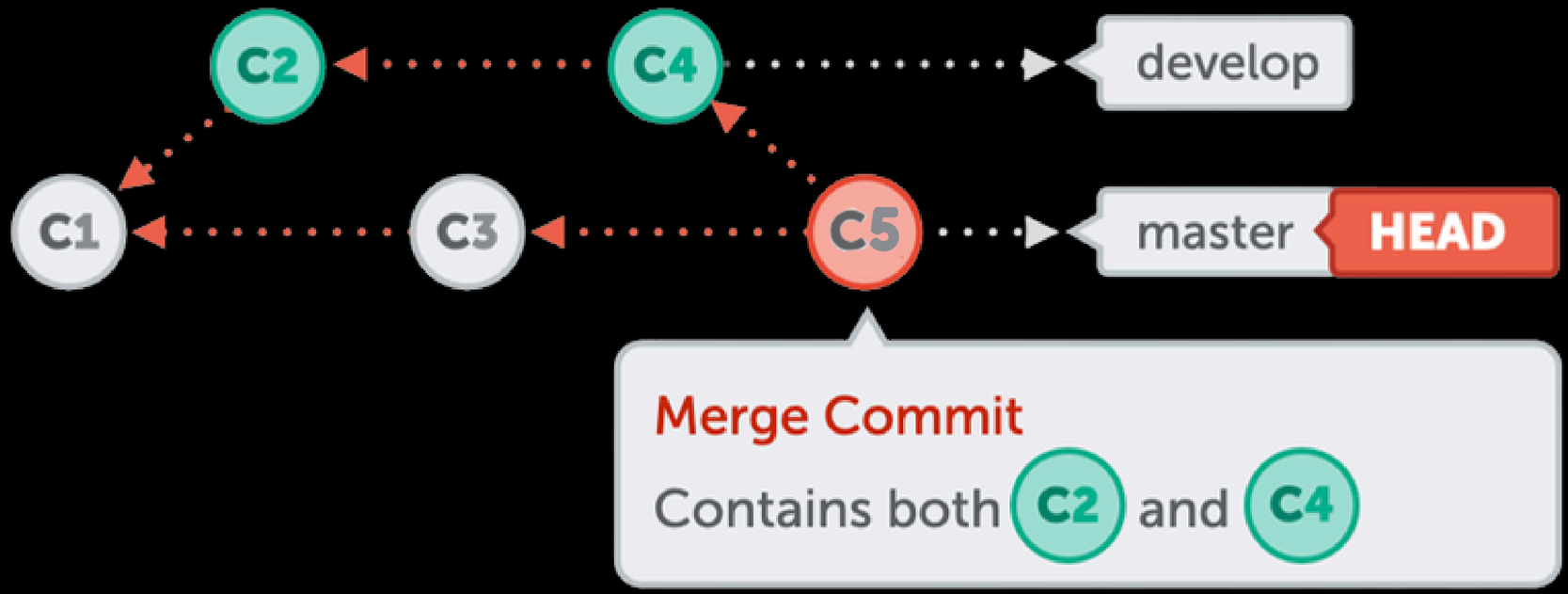
- Cherry picks a Commit

```
$ git cherry-pick <commit-id>
```


C2
C4
develop
C1
C3
C2'
master
HEAD
Cherry-Pick
Only brings in
C2


C2
C4
develop
C1
C3
C5
master
HEAD
Merge Commit
Contains both C2 and C4

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ADVANCE GIT COMMANDS

- Used to create a new commit that undoes the changes made by a previous specific commit

```
$ git revert
```

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- Used to undo changes and modify the commit history

```
$ git reset
```

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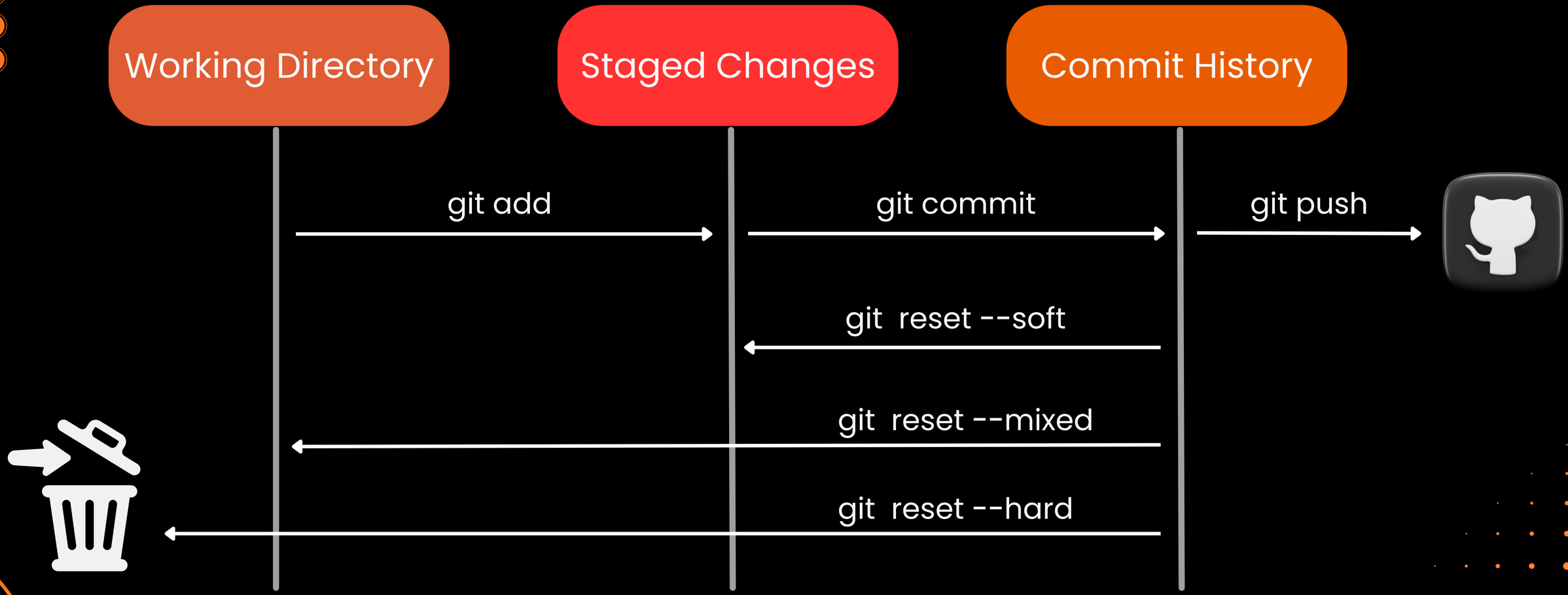

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- Used to move the commits of the current branch on top of any other branch

```
$ git rebase <branch_name>
```

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OPEN SOURCE CONTRIBUTION

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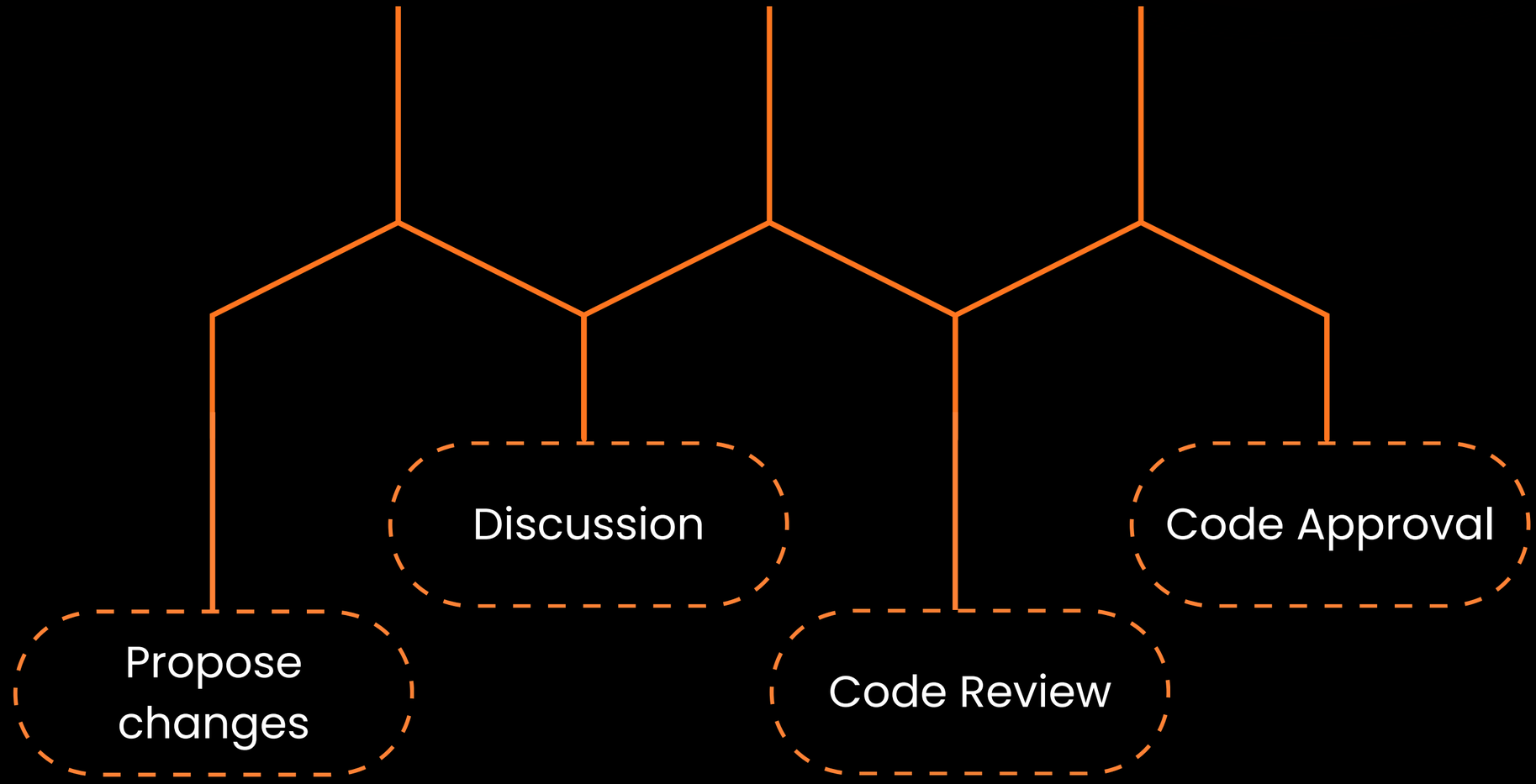
WHAT IS A PULL REQUEST(PR)?



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GETTING STARTED WITH GITHUB!

- Sign up / Sign in

- Go the repository you want to contribute to

- Fork the repository

- Fork the repository

A fork is a copy of the repository on your GitHub

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- Clone your fork and add upstream

  A clone is a local copy of the repository on your computer

- Pull latest changes

  Get the latest changes before you start making changes

```
$ git pull upstream main
```

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Commit your changes

```
$ git add .
$ git commit -m <Commit Message>
```

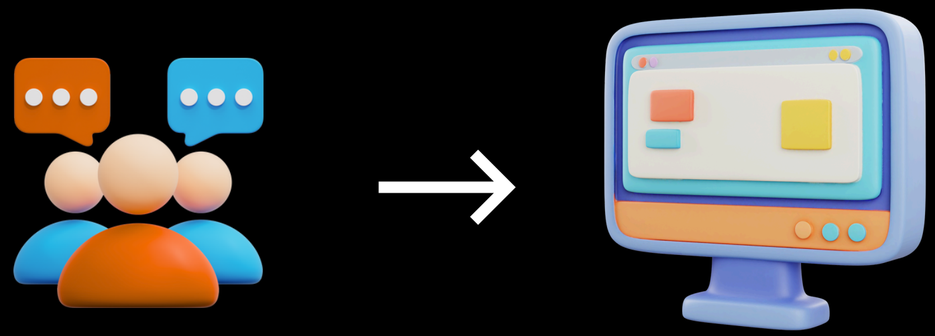
Push your changes

```
$ git push origin <branch_name>
```

- Create a Pull Request (PR)
- Review and Discussion

  Code any changes required, fix bugs, or improve documentation
- Merge the Pull Request

  Once the changes are approved, they will be merged into the main branch



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THANK YOU!

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BREAK TIME